CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 20: Area and Perimeter of Plane Figures

Topic-wise short notes with solved examples

Perimeter and area of triangles (including Heron's formula), quadrilaterals and circles, with problems on paths, rings and combined figures.

1 Triangles

- Perimeter $= a + b + c$. Semi-perimeter $s = \dfrac{a+b+c}{2}$.
- Area $= \dfrac{1}{2} \times$ base $\times$ height.
- **Heron's formula** (all three sides known):

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

Triangle	Area	Other facts
Right-angled, legs a, b	$\frac{1}{2}ab$	Hypotenuse $\sqrt{a^2+b^2}$
Equilateral, side a	$\frac{\sqrt{3}}{4}a^2$	Height $\frac{\sqrt{3}}{2}a$
Isosceles, equal sides a, base b	$\frac{b}{4}\sqrt{4a^2-b^2}$	Height $\frac{1}{2}\sqrt{4a^2-b^2}$

- An altitude can be found from the area: height $= \dfrac{2 \times \text{area}}{\text{base}}$.

Example 1.1 Find the area of a triangle with sides 13 cm, 14 cm and 15 cm, and the altitude on the 14 cm side.

SOLUTION

$s = 21$. Area $= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$ sq cm.

Altitude $= \dfrac{2 \times 84}{14}$

84 sq cm; 12 cm

Example 1.2 The area of an equilateral triangle is $36\sqrt{3}$ sq cm. Find its side and perimeter.

SOLUTION

$\frac{\sqrt{3}}{4}a^2 = 36\sqrt{3}$, so $a^2 = 144$

Side 12 cm; perimeter 36 cm

Example 1.3 An isosceles triangle has equal sides 10 cm and base 12 cm. Find its area.

SOLUTION

Height $= \sqrt{100 - 36} = 8$ cm

48 sq cm

2 Quadrilaterals

Figure	Perimeter	Area
Rectangle, l by b	$2(l + b)$	$l \times b$
Square, side a	$4a$	$a^2 = \frac{1}{2} \times \text{diagonal}^2$
Parallelogram	$2\times$ (sum of adjacent sides)	base $\times$ height
Rhombus	$4\times$ side	$\frac{1}{2}d_1d_2$
Trapezium	Sum of the sides	$\frac{1}{2}(a + b)h$
Any quadrilateral	Sum of the sides	$\frac{1}{2}\times$ diagonal $\times$ (sum of the perpendiculars on it)

- Diagonal of a rectangle $= \sqrt{l^2 + b^2}$. Diagonal of a square $= a\sqrt{2}$.
- A quadrilateral with all sides and one diagonal known: use Heron's formula on the two triangles.

Example 2.1 The diagonal of a square is 12 cm. Find its area and perimeter.

SOLUTION

Area $= \frac{1}{2} \times 144 = 72$ sq cm. Side $= \sqrt{72} = 6\sqrt{2}$ cm.

72 sq cm; $24\sqrt{2}$ cm

Example 2.2 A rectangle has perimeter 46 cm and length 15 cm. Find its area and diagonal.

SOLUTION

Breadth $= 23 - 15 = 8$ cm. Diagonal $= \sqrt{225 + 64}$.

120 sq cm; 17 cm

Example 2.3 The parallel sides of a trapezium are 25 cm and 13 cm, and the non-parallel sides are 10 cm each. Find its area.

SOLUTION

Each end triangle has base $\frac{25 - 13}{2} = 6$ cm and hypotenuse 10 cm, so the height is 8 cm.

$\frac{1}{2} \times 38 \times 8$

152 sq cm

Example 2.4 The area of a rhombus is 216 sq cm and one diagonal is 24 cm. Find the other diagonal and the perimeter.

SOLUTION

$\frac{1}{2} \times 24 \times d = 216$ gives $d = 18$ cm. Side $= \sqrt{12^2 + 9^2} = 15$ cm.

18 cm; 60 cm

Example 2.5 In quadrilateral ABCD, $AB = 9$ cm, $BC = 12$ cm, $CD = 17$ cm, $DA = 8$ cm and $\angle B = 90^\circ$. Find its area.

SOLUTION

$AC = \sqrt{81 + 144} = 15$ cm. Triangle ABC: $\frac{1}{2} \times 9 \times 12 = 54$.

Triangle ACD has sides 15, 17, 8: $8^2 + 15^2 = 17^2$, so it is right-angled, area $\frac{1}{2} \times 8 \times 15 = 60$.

114 sq cm

3 Paths and borders

- Path of width w **outside** a rectangle l by b: area $= (l + 2w)(b + 2w) - lb$.
- Path of width w **inside**: area $= lb - (l - 2w)(b - 2w)$.
- Two cross-paths of width w through the middle: area $= w(l + b - w)$.
- Cost = area × rate per unit area. Number of tiles = area to be covered ÷ area of one tile.

Example 3.1 A lawn 40 m by 25 m has a path 2.5 m wide all around it on the outside. Find the cost of paving the path at ₹80 per sq m.

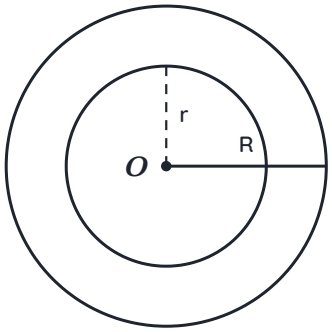
SOLUTION

Outer rectangle 45 m by 30 m: $1350 - 1000 = 350$ sq m

₹28,000

4 Circle

- Circumference $= 2\pi r = \pi d$. Area $= \pi r^2$.
- **Ring** between concentric circles of radii R and r: area $= \pi(R^2 - r^2) = \pi(R + r)(R - r)$.
- Semicircle: area $\frac{1}{2}\pi r^2$, perimeter $\pi r + 2r$. Quadrant: area $\frac{1}{4}\pi r^2$, perimeter $\frac{1}{2}\pi r + 2r$.
- Distance moved by a wheel in one revolution = its circumference.
- Take $\pi = \frac{22}{7}$ unless told otherwise.



A ring: area = π(R² − r²)

Example 4.1 The circumference of a circle is 88 cm. Find its radius and area.

SOLUTION

$2 \times \frac{22}{7} \times r = 88$ gives $r = 14$ cm. Area $= \frac{22}{7} \times 196$.

14 cm; 616 sq cm

Example 4.2 A circular park of radius 21 m has a road 7 m wide around it. Find the area of the road.

SOLUTION

$\frac{22}{7}(28^2 - 21^2) = \frac{22}{7} \times 49 \times 7$

1078 sq m

Example 4.3 A wheel of diameter 70 cm makes 500 revolutions. How far does it travel?

SOLUTION

Circumference $= \frac{22}{7} \times 70 = 220$ cm. Distance $= 500 \times 220$ cm.

1.1 km

Example 4.4 A wire in the shape of a square of side 22 cm is bent into a circle. Find the area of the circle.

SOLUTION

Length of wire $= 88$ cm $= 2\pi r$, so $r = 14$ cm.

616 sq cm

Example 4.5 Find the area of the largest circle that can be cut from a square of side 14 cm, and the area left over.

SOLUTION

Radius 7 cm: circle $= 154$ sq cm. Square $= 196$ sq cm.

154 sq cm; 42 sq cm left

5 Common mistakes

- Using the diameter in place of the radius in πr^2.
- Forgetting the two straight edges in the perimeter of a semicircle or quadrant.
- Adding w once instead of $2w$ to each dimension for a path all round.
- Writing the area of a ring as $\pi(R - r)^2$.
- Mixing units: change everything to the same unit before calculating.
- In Heron's formula, using the perimeter instead of the semi-perimeter.

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